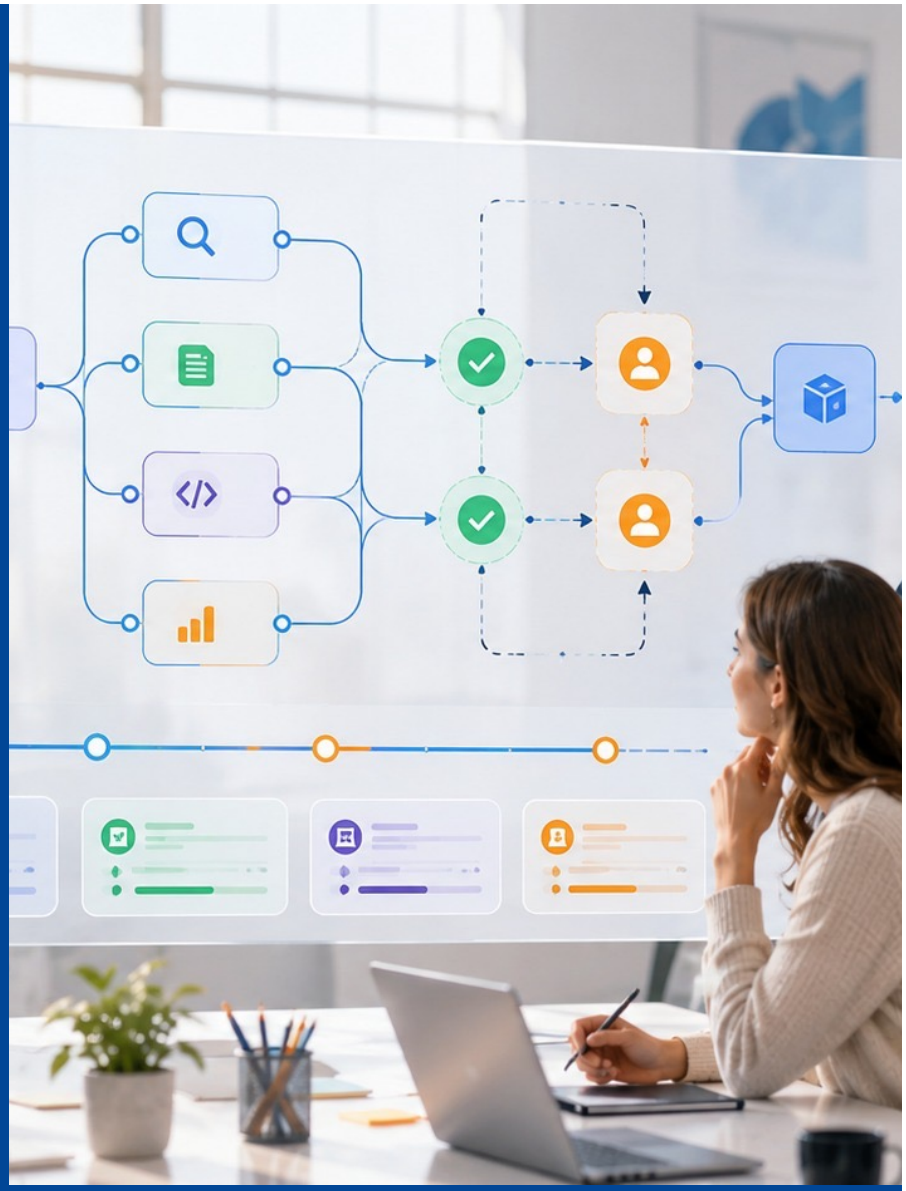
Oversight Is Not a Button

Dr. Yuanting Liu

Head of Human-centered Engineering

fortiss GmbH



Air Canada chatbot

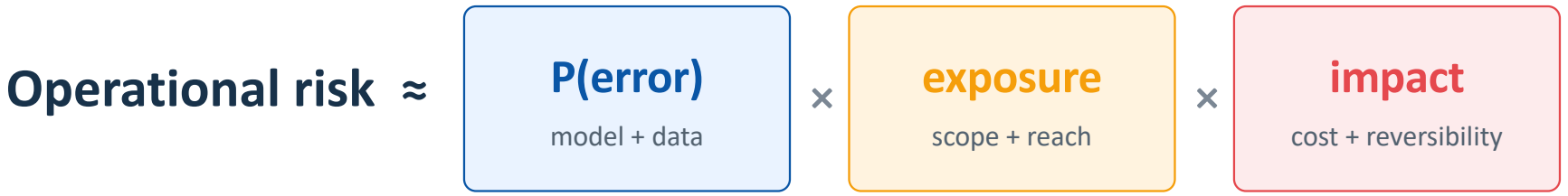
A generated answer became a policy surface.



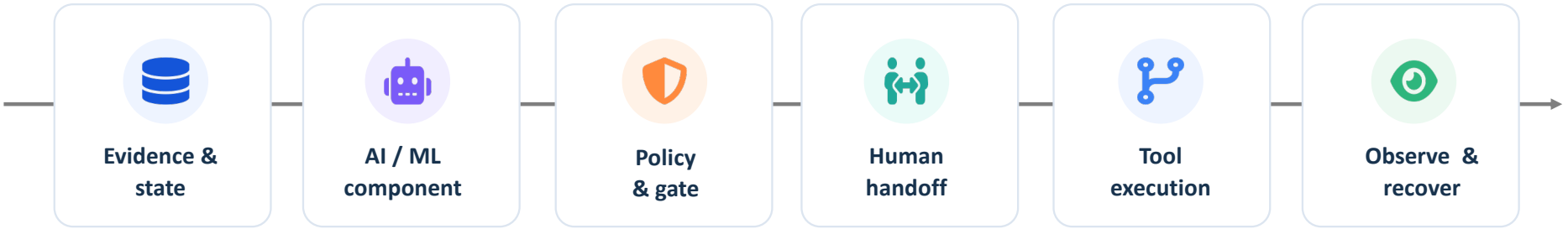
Quelle: https://qz.com/air-canada-ai-chatbot-bereavement-discounts-1851263873

Why oversight is not a nice-to-have

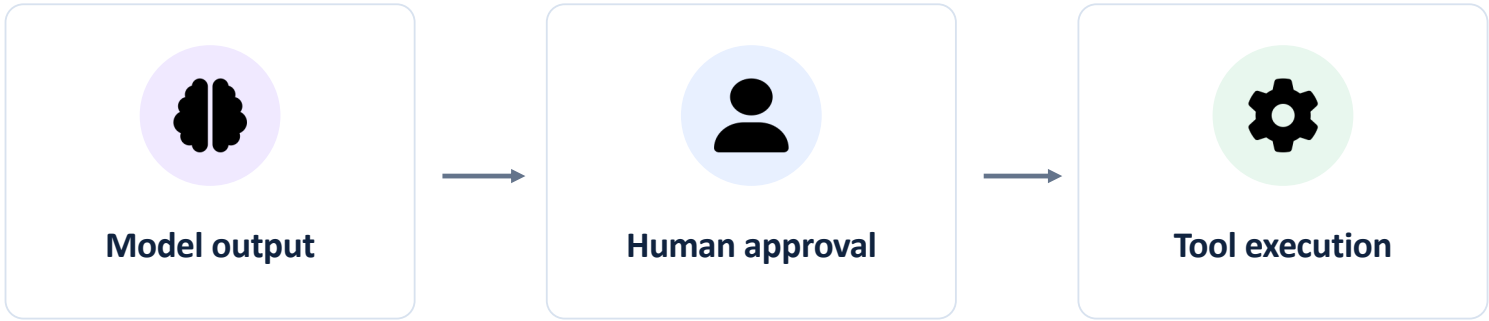
ML reduces error probability. System design controls how far an error can travel.



Oversight is an architecture problem



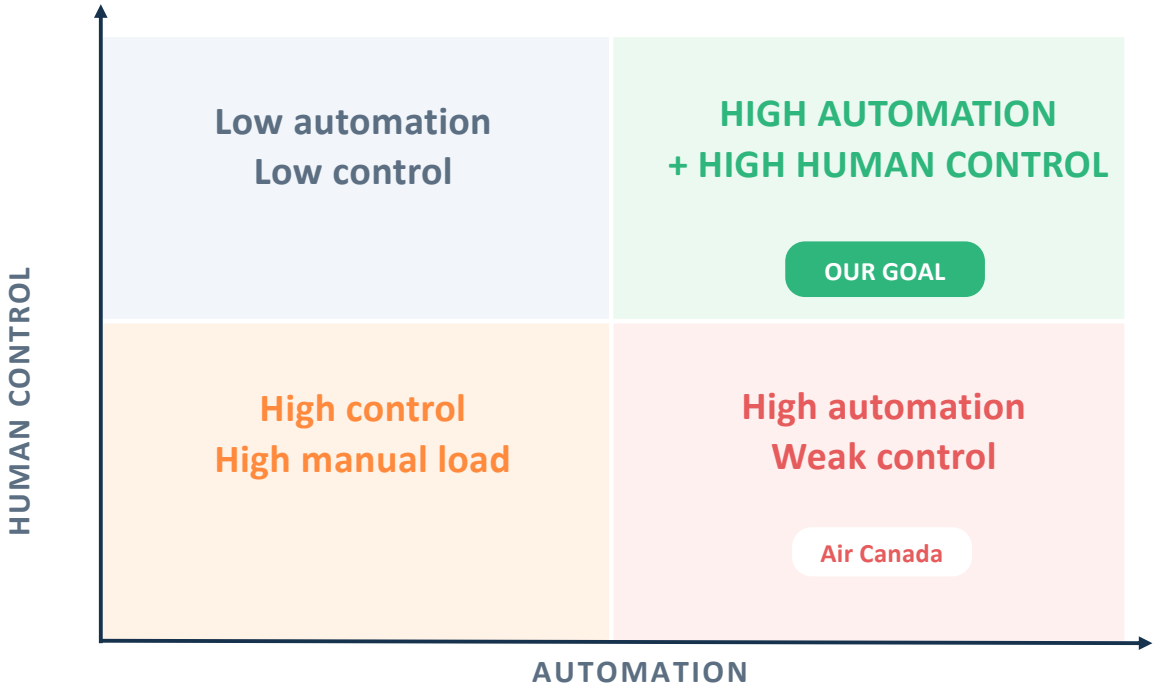
A human in the loop?



≠

meaningful control

High automation & high control



Shneiderman, B. (2020/2022) Human-Centered AI. Oxford University Press.

Oversight at three control points



Before action

MAY IT ACT?

Decision oversight
Evidence · risk · permission

Outcome: act · ask · hand over

During execution

MAY IT CONTINUE?

Execution oversight
Real outcome · scope · stop condition

Outcome: expand · pause · recover

Before update

MAY IT CHANGE?

Update oversight
Meaning · owner · regression risk

Outcome: release · hold · reject

Control point 1: Audit vs. Shape

ANSWER REVIEW

Recommendation-centric

You can request the bereavement discount after travel.

Confidence: 91%

APPROVE AND SEND

Reason backward from the answer

DECISION SHAPING

Process-oriented

GOAL	Determine whether the customer is eligible
EVIDENCE	Current bereavement-fare policy
OPEN GAP	**Has travel already been completed?**
CONFLICT	Chatbot draft conflicts with the policy page
OPTIONS	Explain policy· Ask a question · Hand over

SHOW SOURCE HAND OVER

Reason forward toward a decision

Zhang, Z., Feger, S., Dullenkopf, L., Liao, R., Süsslin, L., Liu, Y., Butz, A. (ACM CSCW 2024). Beyond Recommendations: From Backward to Forward AI Support of Pilots' Decision-Making Process.
Reicherts, L., Zhang, Z., Oswald, E., Liu, Y., Rogers, Y., Hassib, M. (ACM CHI 2025). AI, Help Me Think—but for Myself: Assisting People in Complex Decision-Making by Providing Different Kinds of Cognitive Support.

Control point 2: Crowdstrike outage vs. Canary Deployments



Quelle: https://en.wikipedia.org/wiki/2024_CrowdStrike-related_IT_outages

Control point 2: Crowdstrike outage vs. Canary Deployments



Pre-flight
success signal
stop condition
recovery plan

↓

Canary rollout
small pilot group first

↓

Observe
real outcomes
not only API success

↓

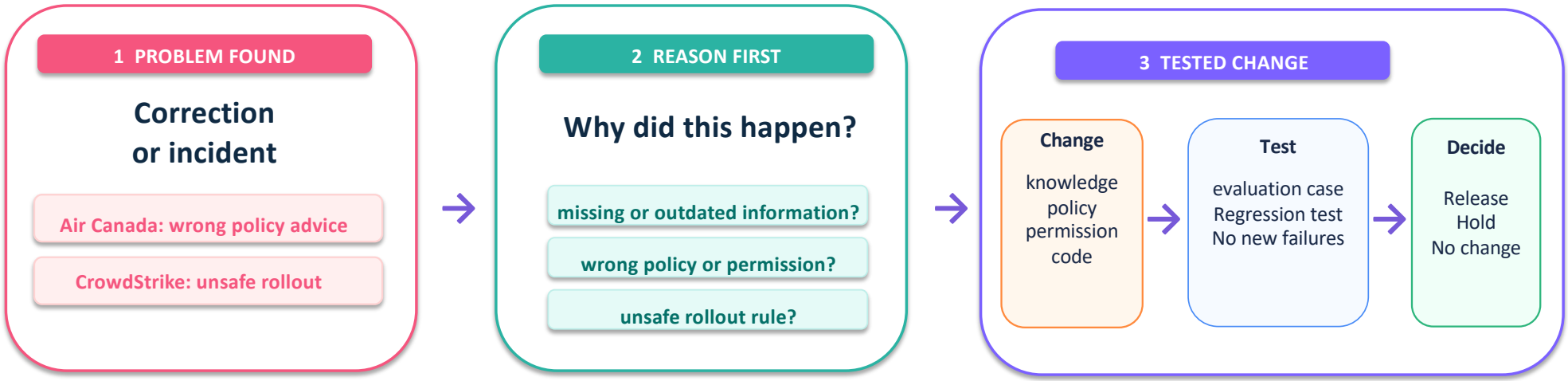
Decide
continue · stop

↓

Recover
rollback or compensate
return to safe sate

Quelle: https://www.crowdstrike.com/en-us/blog/falcon-content-update-preliminary-post-incident-report/

Control point 3: may feedback change the system?



Different incidents, same control questions

Before action
May it act?

Before action
May it continue?

Before update
May it change the system?

AIR CANADA
Answer authority

- Is the policy evidence sufficient for a definite answer?
- Withdraw or correct advice when a conflict appears
- Add policy-source checks and evaluation cases

CROWDSTRIKE
Action reach

- Is the update sufficiently validated for this deployment scope?
- Stop rollout, remediate, and restore systems
- Add validation, staged rollout, and recovery tests

TAKEAWAY

Good oversight is not a brake on useful automation.

- Make the action shapeable
- Make the outcome recoverable
- Make the correction learnable

Thank you · Questions